Context compaction in agent handoffs

Prose summaries and structured checklists

CS3227 Weeks 8 and 9

Five coding histories / 19 October 2026

Handoffs as context engineering

How much useful task context can the next agent recover?

Multi-Agent Coordination

Pass work between agents with different roles and focused context.

Context Engineering

Curate a short handoff from a recorded message history.

Constraint: at most 200 words, including checklist labels

A paired comparison on one repair task

Repair Deskmate's LLM health-check probe

Stage	Input and output	Model
Coding agent	Starting code, contract and tests Recorded message history	Opus 5.5
Handoff agent	Same history for both formats One prose / one checklist	Four models
Receiving agent	Only the handoff and six questions Answers with supporting quotes	Opus 5.5

5 histories × 4 models × 2 formats = 40 handoffs and 40 quizzes

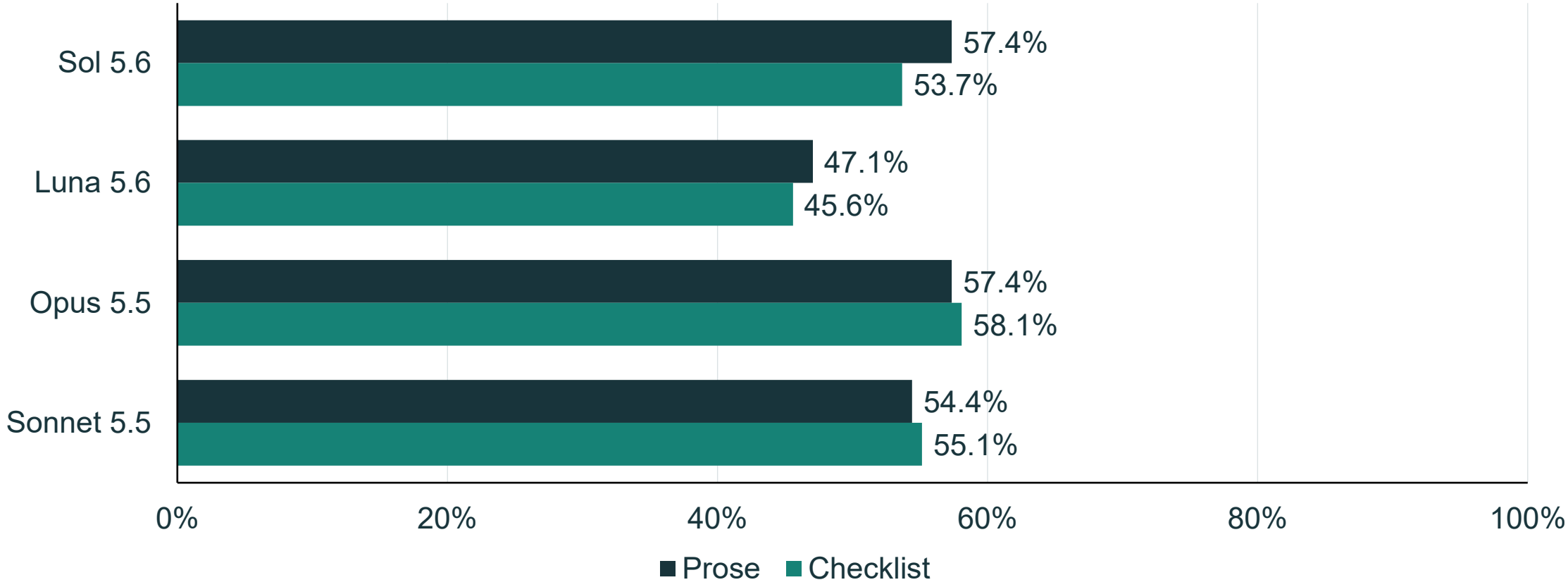
Fact recovery as the assessment

Question topic	Question topic
Q1 Changed files	Q4 Timeout reasoning
Q2 Checks and outcomes	Q5 Successful provider responses
Q3 Verification limits	Q6 Proxy / header behaviour

Recovery rate = correct expected facts / source-supported facts

136 facts per model and format, pooled over five histories
Overall: 583 correct, 505 not stated, 0 wrong expected facts

No consistent checklist advantage



Expected facts recovered (%) / Five histories, 136 facts per bar

Two more histories changed the small leads

Checklist minus prose, in percentage points

Handoff model	Original 3 histories	Expanded 5 histories
Sol 5.6	-6.1 pp	-3.7 pp
Luna 5.6	+2.4 pp	-1.5 pp
Opus 5.5	+2.4 pp	+0.7 pp
Sonnet 5.5	+0.0 pp	+0.7 pp

Positive favours checklist. Luna’s lead reverses.

Exploratory extension after viewing the first three results

One checklist omitted most success criteria

Sol 5.6 checklist, history 04

Handoff excerpt

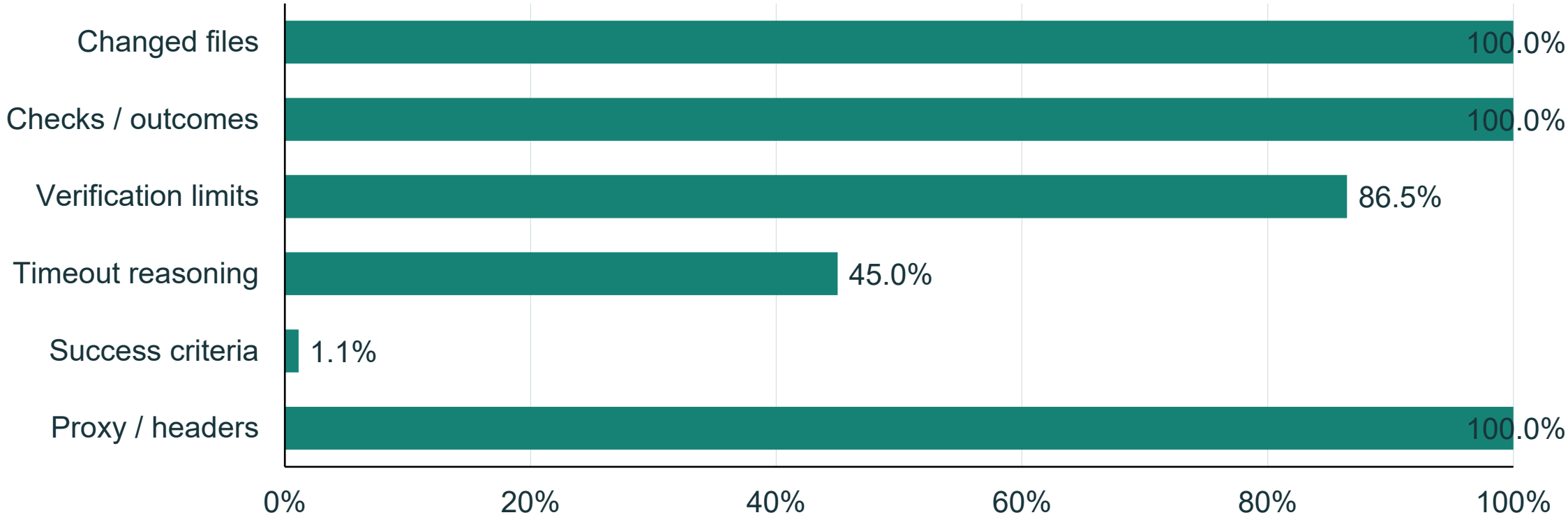
“Offline standalone suite passed all 30 tests in 0.030s.”

Receiver’s Q5 answer excerpt

“not stated. The handoff mentions only response validation and shape validation”

Q5 recovery: 1 of 9 facts, credited from the test-coverage answer

Success criteria were the main omission



4 / 360

success-criteria facts recovered

Different topics contain different numbers of facts.

The word budget limits the comparison

12 / 40 handoffs exceeded 200 words

Handoff model	Pairs with both formats within 200 words
Sol 5.6	5 / 5
Luna 5.6	5 / 5
Opus 5.5	0 / 5
Sonnet 5.5	1 / 5

One task. Five shared histories. Unblinded AI grading.

Recall first in handoff design

Format alone gave no consistent benefit in this setup.

Structured Note Taking

Possible next test: explicitly preserve task success criteria.

Just-in-Time Retrieval

Possible next test: let the receiver retrieve the referenced contract.

These follow-ups remain untested in this experiment.

Backup: complete before-and-after rates

Handoff model	Format	3 histories	5 histories
Sol 5.6	Prose	47/82 (57.3%)	78/136 (57.4%)
Sol 5.6	Checklist	42/82 (51.2%)	73/136 (53.7%)
Luna 5.6	Prose	35/82 (42.7%)	64/136 (47.1%)
Luna 5.6	Checklist	37/82 (45.1%)	62/136 (45.6%)
Opus 5.5	Prose	48/82 (58.5%)	78/136 (57.4%)
Opus 5.5	Checklist	50/82 (61.0%)	79/136 (58.1%)
Sonnet 5.5	Prose	45/82 (54.9%)	74/136 (54.4%)
Sonnet 5.5	Checklist	45/82 (54.9%)	75/136 (55.1%)

Correct facts / source-supported facts, pooled within each condition

Backup: protocol and audit details

Fixed roles: claude-opus-5-5
Handoffs: gpt-5.6-sol, gpt-5.6-luna,
claude-opus-5-5, claude-sonnet-5-5

Fresh sessions and source-based keys frozen before each batch
Same 30 offline tests passed for all five coding repairs

OpenAI native tools stayed exposed under the agreed exception.
Audits found no native tool calls or incomplete final evidence.

Full protocol, exclusions and claim observations: expanded-results/report.html